

A β 42 BindCraft Campaign — Pipeline Summary

2026-05-20

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Date: 2026-05-20

Target: Receptor-bound A β 42 filament, Conformation 1 (PDB 9CO4, chains C/E/G)

Design engine: BindCraft (AF2 backpropagation)

Cluster: Frontenac HPC (A100-SXM4-40GB)

SLURM Job: 8375335 (ran 11 days 1 hour, OOM-killed at 42 GB; resumed as 8415123 with 64 GB)

Internal Pipeline Funnel

BindCraft runs three stages internally in a single integrated loop. The table below shows the equivalent external tool for each stage and the throughput at each level.

Internal Stage	Equivalent To	Count	Notes
AF2 backbone hallucination	RFdiffusion	1,356 started → 924 successful	432 rejected (low pLDDT)
ProteinMPNN seq. design	ProteinMPNN	18,480 sequences	20 per successful trajectory
AF2 complex prediction	ColabFold	18,480 predictions	Every MPNN seq gets AF2 validation
Failed base AF2 filters	—	15,921 (86.2%)	Quick rejection (i_pTM / pLDDT)
Failed advanced filters	—	2,279 (12.3%)	Interface metrics, hydrophobicity, Met
Passed all filters	—	280 (1.5%)	
Final accepted (dedup.)	—	50 designs, 30 scaffolds	Best model per MPNN seq kept

Funnel Visualization

Stage	Description	Surviving
1	Backbone hallucination started	1,356
2	Backbone accepted (pLDDT threshold)	924
3	MPNN sequences generated	18,480
4	AF2 base filter pass	2,559
5	AF2 advanced filter pass	280
6	Final accepted designs	50

Key Rates

Metric	Value
Backbone success rate	68.1% (924 / 1,356)
MPNN → base AF2 pass rate	13.8% (2,559 / 18,480)
Base AF2 → advanced filter pass rate	10.9% (280 / 2,559)
Overall MPNN seq → accepted	0.27% (50 / 18,480)
Overall trajectory → accepted design	5.4% (50 / 924)
Unique scaffolds	30
Designs per scaffold	1-2

Accepted Design Quality (n = 50)

Metric	Min	Mean	Median	Max
i_pTM	0.56	0.75	0.76	0.86
pTM	0.72	0.81	0.82	0.89
pLDDT	0.81	0.90	0.91	0.94
dG (kcal/mol)	-98.9	-76.1	-76.5	-49.2
Length (aa)	60	77	79	90
Interface residues	14	27	26	40
H-bonds	3.0	5.3	4.5	11.5
MPNN recovery	0.21	0.39	0.38	0.64
Surface hydrophobicity	0.22	0.31	0.32	0.35

Design Parameters

Parameter	Value
Target PDB	9CO4 chains C, E, G
Hotspots	Y10, E11, H13, H14, Q15, K16 × 3 chains = 18 positions
Binder length	60-90 aa (uniform sampling)
Design algorithm	4stage
MPNN	Soluble weights, 20 seqs, fix interface, temp 0.1
Omit AAs	Cysteine
Target accepted designs	2 1,000 (run until reached or wall time)